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HW ①

1) $v(t) = 2\cos(\omega t + 30)$

$Z = 5 + j10$

Find I and Pave

$V = 2 \angle 30°$

$Z = 5 + j10$

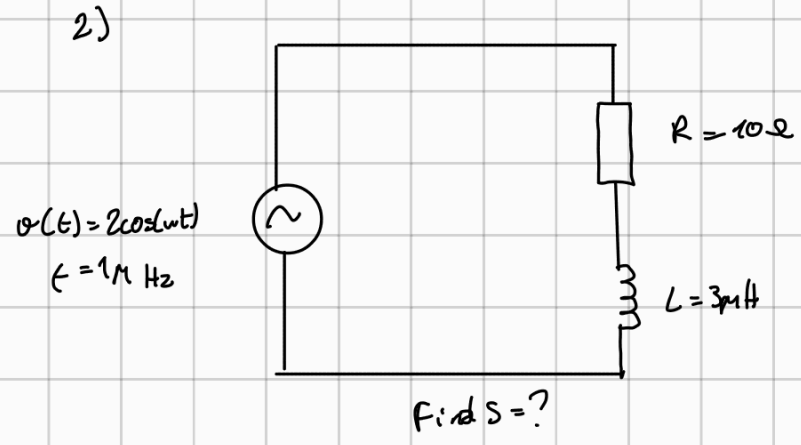
$V = I \cdot Z \Rightarrow I = \frac{V}{Z} = 0,1789 \angle -2,52°$

$S = \frac{1}{2} V \cdot I^* = 0,1789 \angle 1,107°$

$= 0,1704 - j0,0543$

$P = Re\{S\} = 0,1704\ W$ //

2)



$f = 10^6\ Hz \Rightarrow \omega = 2\pi \cdot 10^6\ Hz$

$R = 10\Omega \Rightarrow Z_R = 10\Omega$

$L = 3\mu H = 3 \cdot 10^{-6}\ H \Rightarrow Z_L = j\omega L = j\, 2\pi \cdot 10^6 \cdot 3 \cdot 10^{-6} = j6\pi$

$V = 2 \angle 0°$

$V = I \cdot Z_{eq}$ $\qquad Z_{eq} = Z_R + Z_L = 10 + j6\pi$

$I = \frac{V}{Z_{eq}} = \frac{2 \angle 0°}{10 + j6\pi} = 0,0937 \angle -1,083°$

$S = \frac{1}{2} V \cdot I^* = 0,0937 \angle 1,083° = 0,0439 + j0,0828$